NetC14: A Gaussian mixture neural network for calibrating and summarizing radiocarbon dates

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Abstract
We propose NetC14, a quasi-parametric Gaussian mixture neural network for calibrating and summarizing radiocarbon dates using regularized Maximum Likelihood. The model serves as a proof of concept for neural networks applied to the problem of radiocarbon dating. It is evaluated using simulated data and compared to two traditional approaches (Summed Probability Distribution and Composite Kernel Density Estimation) as well as two recently proposed Bayesian mixture models fit with MCMC (Bayesian Gaussian Mixture Model and Dirichlet Process Mixture Model). All models are further applied to two published archaeological datasets from Iron Age Ireland and Paleoindian North America. Results indicate that NetC14 attains parity with existing mixture models on both age and density estimation while achieving substantially lower compute times and superior scaling, even absent GPU acceleration. NetC14 also inherits the extensibility of neural networks, allowing for integration of additional diagnostic data, debiasing techniques, and demographic constraints.

Keywords: Deep Learning, SparseMax, Regularized Maximum Likelihood, Density Estimation, Paleodemography, Bayesian Inference

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1. Introduction

Archaeological radiocarbon (^{14}C) dating has two primary objectives: (i) to estimate the true ages of a set of organic samples and (ii) to reconstruct the true calendar age distribution $p(t)$ from which those samples are drawn, with the latter serving as a proxy for population size over time. The first objective was established by the chemist Willard Libby in the late 1940s (Libby, 1946), the second by the archaeologist John Rick four decades later (Rick, 1987). Among archaeologists, a common strategy for achieving them is to first estimate the ages of samples individually and only then bring them together to reconstruct the distribution from which the samples are drawn, typically by summing the posterior probabilities of the individual age estimates, the Summed Probability Distribution (SPD) method (Blockley et al., 2000; Crema, 2022). Unfortunately, SPD estimates are noisy (Brown, 2015) and statistically incoherent, summing posteriors with contradictory priors (Carleton & Groucutt, 2021; Crema, 2022; Heaton, 2022).

Recently, however, Heaton (2022) and Price et al. (2021) have proposed an alternative Bayesian strategy, pursuing both radiocarbon dating objectives simultaneously, estimating ages and the chronology they are drawn from at the same time using latent variable models. To the extent that their approaches differ, it is in the specification of $p(t)$. Price et al. (2021) recommend a parametric Bayesian Gaussian Mixture Model (BGMM), Heaton (2022) a non-parameteric Dirichlet Process Mixture Model (DPMM), with the latter learning the number of Gaussian mixture components during model training. Here we propose a third mixture model, a Gaussian Mixture Neural Network that we call NetC14. NetC14 is quasi-parametric as it begins with an initially over-specified number of Gaussian components, then uses a SparseMax activation (Martins & Astudillo, 2016) on mixture weights to prune components out of the model.

As a proof of concept, we evaluate the ability of NetC14 to calibrate and summarize radiocarbon dates using simulated data, comparing its performance to that of the other mixture models - BGMM and DPMM - plus two traditional approaches - SPD and Composite Kernel Density Estimation (CKDE). We also apply all models to real datasets used in Heaton (2022) and Bronk Ramsey (2017). Our results show that a minimal specification of NetC14 - severely limited in its complexity and implementation - is already competitive with the state of the art, but substantially more computationally efficient and scalable to larger datasets, and that is before harnessing GPU acceleration and other optimizations. NetC14 also comes with all the advantages of neural

networks, making it highly flexible and adaptable to a wide variety of data, archaeological and otherwise, which we discuss below.

2. Background

In this section, we review the Bayesian logic of the mixture models. For more background on archaeological radiocarbon dating, see (Bayliss, 2009; Bronk Ramsey, 2008; Hajdas et al., 2021; Taylor & Bar-Yosef, 2016; Wood, 2015). For detail on how these probabilities are estimated with matrix algebra in R, see Supplement A (Introduction).

2.1. Generative Model

Our data consist of fraction modern radiocarbon measurements, a dimensionless unit comparing the $^{14}C/^{12}C$ ratio in organic samples to a modern reference standard of atmospheric carbon levels in 1950 (McNichol et al., 2001). Adapting Heaton (2022) to this fraction modern context, we define the generative model for radiocarbon measurements F_i on $i = 1, ..., n$ organic samples as

$$\begin{aligned} t_1, ..., t_n &\sim p(t) \\ F_i &\sim N(c(t_i), z_i^2) \end{aligned} \tag{1}$$

where $t_1, ..., t_n$ are the true but unknown ages of the n samples drawn from the age distribution $p(t)$, $c(t)$ is the fraction modern predicted by a radiocarbon calibration curve for a given age t, and z_i is the standard deviation of the measured fraction modern for sample i. In plain English, we might think of this generative model as answering two archaeological questions. First, how likely is it that someone in the past did something at a particular time and place that leaves some organic material behind, like butchering a deer or chopping down a tree? That is a random draw of a calendar age t_i from $p(t)$. Second, assuming that organic material preserves over the intervening years, how much ^{14}C should we expect to find in it relative to a modern reference standard when we actually get around to measuring it? That observation is a random draw of a fraction modern measurement F_i from $N(c(t_i), z_i^2)$, a Gaussian measurement model that incorporates uncertainty from both the measurements and the calibration curve.

We denote uncertainty in the calibration curve itself for a given age t as

$$c(t) \mid t \sim N(\mu_c(t), \sigma_c(t)^2) \tag{2}$$

with $\mu_c(t)$ and $\sigma_c(t)$ being the point-wise means and standard deviations of Gaussians defined by the calibration curve. Conventionally, these quantities are expressed as ^{14}C ages, but we assume that they have been converted to fraction modern using the mean-life function for radiocarbon. Given that F_i are also sampled from a Gaussian, the curve can be marginalized over, such that

$$F_i \mid t_i \sim N(\mu_c(t_i), \sigma_c(t_i)^2 + z_i^2) \tag{3}$$

which shows that uncertainty propagates through the calibration curve and into the measurements.

2.2. Bayesian Mixture Models

The mixture models evaluated in this paper seek to reconstruct the calendar age distribution $p(t)$ by conditioning on the mixture parameters $\theta = \{w_k, \mu_k, \sigma_k\}$ (the weights, means, and standard deviations) of $k = 1, ..., K$ clusters or component distributions f

$$p(t \mid \theta) = \sum_{k=1}^{K} w_k \, f(t \mid \mu_k, \sigma_k^2) \tag{4}$$

where the weights must sum to one and f is typically specified as a Gaussian. The use of mixture models is interesting here for a number of reasons. First, they are universal approximators, capable of reconstructing any continuous density function provided they have a sufficient number of component distributions, though naturally that comes with the risk of over-fitting (Li & Barron, 1999; McLachlan & Peel, 2000; McLachlan et al., 2019). They are relatively straightforward to grasp, too, especially if one already understands how SPD models are fit, as they both involve summing probability distributions. Finally, the latent clustering assignments are open to a great deal of archaeological interpretation - or over-interpretation as the case may be.

In a Bayesian context, we aim to estimate the mixture parameters θ by sampling from their posterior probability given the observed fraction modern measurements F, that is

$$p(\theta \mid F) \propto p(F \mid \theta) \, p(\theta) \tag{5}$$

with $p(F \mid \theta)$ being the likelihood of the measurements given the mixture parameters, what we refer to as the *mixture likelihood*, and $p(\theta)$ the prior probability of the parameters. To sample from this posterior, we need to (i) define the mixture likelihood $p(F \mid \theta)$ and (ii) specify the priors $p(\theta)$

for the set of weights, means, and standard deviations of the mixture model. We will elaborate on the priors in Heaton (2022) and Price et al. (2021) in the methods section. For now, the likelihood function warrants some attention.

To define the likelihood, we first define the joint probability of an observed fraction modern measurement F_i and its calendar age t_i as

$$p(F_i, t_i \mid \theta) = p(F_i \mid t_i)\, p(t_i \mid \theta) \tag{6}$$

where $p(F_i \mid t_i) = N(F_i \mid \mu_c(t_i), \sigma_c(t_i)^2 + z_i^2)$ is the likelihood of a measurement conditioned on its calendar age, what we refer to as the *measurement likelihood*, and $p(t_i \mid \theta)$ is the mixture density estimated by the mixture model at t_i. This encapsulates the generative model, weighting the probability of obtaining measurements F_i with ages t_i by the prior probability of those ages. Note that we drop θ from the measurement likelihood since once we know a sample's age, the mixture tells us nothing more about its measurement. We also assume the measurements are conditionally independent given their ages, ignoring correlation induced by marginalizing over the shared calibration curve.

Because we do not know the true ages, however, we consider all possible ages instead, thus evaluating the likelihood for sample i as

$$p(F_i \mid \theta) = \int p(F_i \mid t)\, p(t \mid \theta)\, dt \tag{7}$$

As Price et al. (2021) show, this integral can be approximated using Riemann sums. One first evaluates the measurement likelihood $p(F_i \mid t)$ over an annually resolved grid of m years. Repeating this procedure for each of the n samples results in an $n \times m$ measurement matrix $\mathbf{M}$ whose rows are measurement likelihoods for the samples given the calibration curve. One then evaluates the mixture density $p(t \mid \theta)$ over the same grid of m years to produce a length-m column-vector $\mathbf{v}$. Taking the matrix product

$$\mathbf{h} = \mathbf{M}\mathbf{v} \tag{8}$$

(equation 14 in Price et al. (2021)) gives the vector of sample likelihoods $\mathbf{h}$, where the product of its elements is the total likelihood (the sum of their logs being the log-likelihood). This compact expression shows that just two data structures are needed to estimate the full mixture likelihood: a

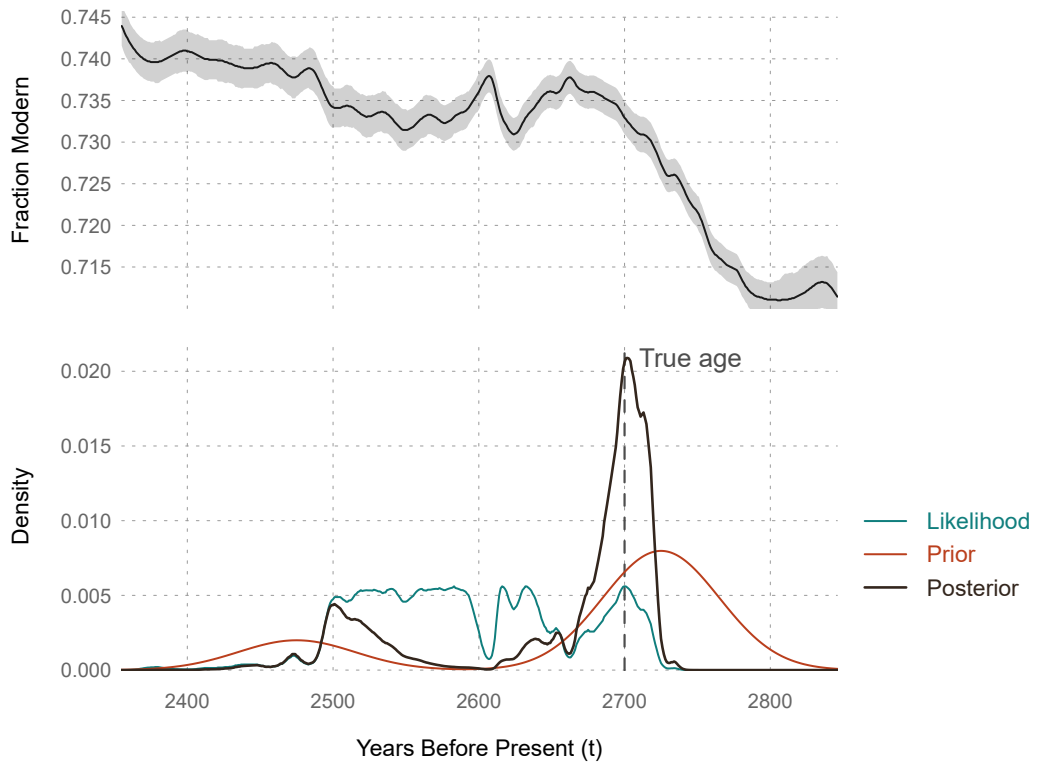


Figure 1: Radiocarbon calibration with measurement likelihood $p(F \mid t)$ weighted by the mixture density $p(t \mid \theta)$, which acts as an informative prior concentrating posterior probability mass near modes of the mixture density and also, as a consequence, around the true age of the sample. The calibration curve is plotted above with a gray ribbon for the 95% confidence interval.

matrix $\mathbf{M}$ of measurement likelihoods and a vector $\mathbf{v}$ of probability densities. Any model that can estimate $\mathbf{v}$ can, thus, be fit using Maximum Likelihood (e.g., Timpson et al. (2020)).

2.3. Posterior Age Estimation

According to Bayes' Theorem, the probability density of the posterior age distribution is

$$p(t \mid F, \theta) = \frac{p(F \mid t)\, p(t \mid \theta)}{p(F \mid \theta)} \tag{9}$$

In words, we weight the measurement likelihood for each sample by the mixture distribution and then normalize with the mixture likelihood. This is equivalent to multiplying each row of $\mathbf{M}$ elementwise by $\mathbf{v}$, and then normalizing each row to sum to one (Supplement A, Example). Weighting in that way has the effect of pulling the probability mass of each sample toward the modes of the mixture density, as shown in Figure 1, with the consequence that scalar estimates for

each sample will be drawn more often from those periods, thus providing a Bayesian means of overcoming multi-modality in the calibration curve.

3. Methods

For this analysis, we rely on the IntCal20 calibration curve (Heaton et al., 2020; Reimer et al., 2020). The analysis is conducted in the R programming language and environment, v4.5.0 (R Core Team, 2025), with the actual analysis requiring just three packages: `torch` (Falbel & Luraschi, 2026), `carbondate` (Heaton et al., 2024; Heaton, 2022), and `parallel`. Computations are run on a personal computer with Windows 11, using a 12th Gen Intel(R) Core i9-12900KF with 16 cores and 64 GB RAM. Code and data can be found at the Zenodo archive (Vernon, 2026) and at the GitHub repository *kbvernon/netc14*.

3.1. Simulations

To compare models, we rely on both simulated and empirical data. Each simulation requires that we specify four variables: the number of clusters K, the kind of distribution f, the number of samples n, and the number of years m over which to simulate a discretized calendar age distribution $\mathbf{v}$. Their possible values are $K = \{1, 3, 9\}$, $f = \{\text{Gaussian, Uniform, Skewed}\}$, $m = \{500, 2500, 12500\}$, and $n = \{50, 250, 1250\}$. The three distributions represent clusters that are symmetric, flat, and asymmetric, the last rising quickly and falling slowly or the reverse (the Skewed distribution is a Gamma reflected at random). We choose these values to achieve a range well beyond what previous studies have attempted, making tests considerably more challenging. Every combination of these variables is evaluated resulting in $3^4 = 81$ unique starting conditions or *scenarios*. For each scenario, a start date t_{min} is randomly sampled from a discrete uniform on the interval from 0 to $(55000 - m)$ years before present (ybp, with present = 1950) so that the end date $t_{max} = t_{min} + m - 1$ never exceeds the upper limit on radiocarbon dating. All 81 scenarios are repeated 28 times, each with a unique start date, resulting in 2268 total simulations spread out randomly across the viable range of radiocarbon dating (Supplement A, Simulations).

A single simulation provided with one of those scenarios and starting years consists of two stages. In the first, we randomly generate a calendar age distribution $p(t)$. In the second, we implement the generative model. To simulate $p(t)$, we randomly sample the parameters $\theta = \{w_k, \mu_k, \sigma_k\}$ for a mixture of K Gaussians using

$$
\begin{aligned}
w_k &\sim \text{Dirichlet}(\mathbf{1}_K) \\
\mu_k &\sim \text{DiscreteUniform}(B_k) \\
\sigma_k &\sim \text{Gamma}(4, 4/s)
\end{aligned}
\tag{10}
$$

where B_k is the kth of K equally spaced bins partitioning $[t_{min}, t_{max}]$ and $s = m/20$ is the scale of the standard deviations. The discrete uniform encourages separation between components and multi-modality in the full mixture distribution. The scale parameter ensures that component standard deviations have a mean of s, which scales with the width of the interval m, and a 95% interval of $[0.27s, 2.19s]$. For example, a simulation covering 500 years would generate distributions with expected standard deviation of 500/20 = 25 years and a 95% interval of [6.81, 54.80] years. For 2500 and 12500 year intervals, that increases to 125 [34.06, 273.98] and 625 [170.29, 1369.89] years, respectively. The resulting mixture is then evaluated over a discrete, annually-resolved grid in the interval $[t_{min}, t_{max}]$ and truncated to remove any probability mass outside of that interval, giving us the length m vector $\mathbf{v}$.

The only remaining quantity to simulate is the measurement standard deviation z_i, which we sample first in ^{14}C years from a $\text{Gamma}(2, 1/3)$, right-shift by 20 to obtain a mean measurement error of 26-years, then convert to fraction modern using the mean-life equation for radiocarbon. The generative model is then implemented to obtain measurements F_i, which are calibrated in turn with IntCal20 within the m length interval $[t_{min}, t_{max}]$, thus forming the $n \times m$ measurement matrix $\mathbf{M}$, the core data structure used by all models except the DPMM, which takes the uncalibrated fraction modern measurements F and errors z directly.

3.2. Models

Implementation details for each of the following models, including the R code, can be found in Supplement A (Models).

3.2.1. Summed Probability Distribution (SPD)

The SPD model works directly on the measurement matrix $\mathbf{M}$, normalizing each row to obtain individual posteriors, then taking the column means. The result is a noisy estimate $\hat{\mathbf{v}}$ of $\mathbf{v}$. Drawing scalar age estimates $\hat{t}_1, ..., \hat{t}_n$ with SPD involves sampling from the posteriors in the normalized rows of $\mathbf{M}$.

3.2.2. Composite Kernel Density Estimation (CKDE)

There are many versions of KDE for radiocarbon dating (Bronk Ramsey, 2017; McLaughlin, 2019). The version tested here is similar to the one implemented in `rcarbon` and described in Brown (2017). It is an SPD convolved with a Gaussian kernel in the sense that it fits n_{iter} KDE to n_{iter} age samples drawn from each normalized row of $\mathbf{M}$ and then averages them to obtain the estimate $\hat{\mathbf{v}}$, with confidence intervals obtained in a similar way. For age estimation, we use the Bayesian posterior age sampling approach described above, multiplying $\hat{\mathbf{v}}$ as a prior through the rows of $\mathbf{M}$ to obtain posterior age estimates, then inverse transform sampling from those posteriors to get $\hat{t}_1, ..., \hat{t}_n$, so this approach shares information across samples indirectly. Note that we rely on R's defaults for KDE, using a bandwidth determined by Silverman's Rule.

3.2.3. Bayesian Gaussian Mixture Model (BGMM)

Our version of the parametric BGMM follows Price et al. (2021), with a few changes to make it more flexible and faster. To choose the number of mixture components, Price et al. (2021) sample the full posterior for each candidate K and select the one with the highest expected log predictive density, estimated by Pareto-smoothed importance sampling leave-one-out cross-validation [PSIS-LOO; Vehtari et al. (2017)]. To avoid sampling every candidate, for each K we place the component means at evenly spaced quantiles of the samples' expected calendar ages, which puts more components where calendar ages are clustered. We then approximate the posterior mode of θ using 500 iterations of the Adam optimizer in Torch (Kingma & Ba, 2017), and select the K with the lowest BIC, which balances how well the mixture fits the dates against how many components it uses.

The mixture weights have a symmetric Dirichlet prior with concentration $\alpha = 1$. As in Price et al. (2021), the prior on the mixture means is flat, and the means are constrained to be ordered to prevent label switching. We also use a Gamma prior on the mixture standard deviations, but rather than set its shape and rate directly as in Price et al. (2021), we fix its mean at m/K, so the prior scales with the width of the interval and the number of components, and we let a coefficient of variation control the spread around that mean. The default is 0.6, which makes it a weak prior.

The posterior is sampled with 4,000 iterations of Hamiltonian Monte Carlo implemented in TorchScript, with a step size and dense mass matrix tuned during burn-in (Hoffman & Gelman, 2014). Price et al. (2021) start every chain from the same random draw, but we instead start from

the posterior mode of the selected model, slightly jittered, which helps us test for convergence (Supplement A, Convergence Test). Each retained draw $\theta^{(s)}$ defines a mixture that we evaluate over the year grid. The density estimate is the average of these, $\hat{\mathbf{v}} = 1/n_s \sum \hat{\mathbf{v}}^{(s)}$, and credible intervals are obtained the same way. Age estimates are obtained using the posterior age estimation approach described above.

3.2.4. Dirichlet Process Mixture Model (DPMM)

The non-parametric DPMM proposed by Heaton (2022) learns the number of components K using a Gibbs sampler, which works by sampling each unknown quantity in turn while holding the others fixed. The key quantity is a latent clustering variable c_i that records which cluster each radiocarbon sample i belongs to. These assignments are drawn using Pólya urn sampling (Neal, 2000). In plain English, a ^{14}C sample i joins an existing cluster with probability equal to how many samples the cluster currently holds and how well its current age estimate t_i fits that *specific* cluster, and it joins a new (currently empty) cluster with probability equal to a concentration parameter α_d weighted by how well its age fits an *average* cluster. Because samples are reassigned in every iteration, clusters can empty out and new ones can form, so the number of clusters can shrink or grow.

The means and precisions (inverse variances) of the clusters share a Normal-Gamma prior, which is updated each iteration using the samples currently assigned to each cluster. More samples in a cluster raise its expected precision, while more spread in their ages lowers it. Precision is also penalized when a cluster's mean drifts away from the overall mean. Given its cluster assignment, each sample's age is then updated by weighting its measurement likelihood $p(F_i \mid t)$ by that cluster's Gaussian, rather than by the full mixture density, with draws obtained by slice sampling (Neal, 2003). So, every iteration gives each sample one age estimate. Because samples wander across clusters in proportion to their likelihood, these per-iteration ages are analogous to the mixture-weighted posterior age estimates described above.

The Gibbs sampler is implemented in performant C++ code in the R package `carbondate`. The default of 100,000 iterations is impractical given the number of simulations, so we limit sampling to 50,000. The density estimate $\hat{\mathbf{v}}$ is the predicted calendar age distribution for a new sample, averaged over a subset of iterations. In each iteration, that prediction mixes the occupied clusters, weighted by their sizes, with a small chance of a new cluster, weighted by α_d.

3.2.5. Gaussian Mixture Neural Network (NetC14)

NetC14 estimates the mixture parameters θ by minimizing a penalized negative log likelihood

$$J(\theta) = -\ell(\theta) + \gamma\kappa(\theta) \tag{11}$$

where $\ell(\theta) = \sum_{i=1}^{n} \log \mathrm{h}_i(\theta)$ is the log likelihood, h_i is the ith element of the sample likelihoods $\mathbf{h}$ described above, and $\kappa(\theta) = m \sum \sigma_k^{-2}$ is a penalty on the precision of the active components ($w_k > 0$), weighted by γ. The first term rewards mixtures that make the observed measurements likely. The second keeps any one component from collapsing onto a handful of dates, a well-known degeneracy in mixture models (Day, 1969), with m acting as a normalizing constant so that γ is comparable across intervals of different length. Minimizing $J(\theta)$ is equivalent to finding the *Maximum A Posteriori* (MAP) estimate of $p(\theta \mid F)$ under an exponential prior on the component precisions, or equivalently an inverse-gamma prior on the variances (Chen et al., 2008; Ciuperca et al., 2003). The precision penalty puts a floor on the variance of a cluster at roughly $2\gamma m/n_k$, where n_k is the number of dates assigned to it. Below that point, the likelihood can no longer make up for the penalty, so narrow clusters can be recovered only if they are well populated. NetC14 can, therefore, be considered a Bayesian model, but rather than sampling the full posterior, it finds the single most probable set of parameters.

We initialize NetC14 with an overspecified number of components using $K_0 = \lceil 2\sqrt[3]{n} \rceil$, a modification of the Scott rule for optimal bin counts in histograms that allows the initial component count to grow with the number of samples to 8, 14, and 22 for $n = \{50, 250, 1250\}$ (Scott, 1979). The components are then initialized with equal weights, with means at the centers of K_0 equal-width intervals spanning the grid and standard deviations equal to half those interval widths. As in the BGMM, the means are not penalized but are ordered and constrained to the interval, the standard deviations are kept positive, and the weights are passed through a SparseMax activation (Martins & Astudillo, 2016), which, unlike the more familiar SoftMax, can set weights exactly to zero. The activation works by centering the logits around a threshold and then applying ReLU, with the threshold determined so that positive centered logits sum to one and the rest are set to zero. Pruning components in this way rather than adding them satisfies the requirement imposed by neural networks that the dimensions of the data be known in advance. Importantly, because the precision penalty only applies to active components, it also favors sparsity in the mixture.

To get a rough sense of uncertainty, Gaussian noise with fixed scale $\sigma_\epsilon = 0.02$ is added to the unconstrained parameters during training, so the optimizer minimizes the average loss over small random perturbations of θ, that is, $J(\theta + \epsilon)$ with $\epsilon \sim N(0, \sigma_\epsilon^2)$, which amounts to fixed-variance variational inference (Graves, 2011). The density estimate is then obtained by running $n_s = 256$ noisy forward passes and taking their pointwise median, with the 2.5% and 97.5% quantiles giving approximate 95% intervals, though not strictly credible ones. Age estimates are obtained using the posterior age estimation approach described above, with the fitted density $\hat{\mathbf{v}}$ as a plug-in prior. Because $\hat{\mathbf{v}}$ is a single point estimate, these posteriors do not average over uncertainty in θ as a fully Bayesian model would, so they may be overconfident. The noisy forward passes do not correct for this, either, as they have no strictly probabilistic interpretation. We evaluate the practical consequences of this using a proper scoring rule (see below).

NetC14 includes a temperature π that encourages pruning by SparseMax and is set to $\pi = 0.6$ by default. The penalty weight γ is then chosen by five-fold cross-validation on the held-out likelihood of simulated radiocarbon measurements. Because this criterion depends only on the observed measurements and the calibration curve, not on the true ages or density, it could in principle be computed for any dataset, including the real-world examples below. To set a default, we apply it to a panel of simulated chronologies generated independently of those used for evaluation, and choose the value with the smallest median held-out loss relative to the best value for each chronology, giving $\gamma = 16$ (Supplement B). The model is implemented as a single layer torch module and trained over 512 epochs using the Adam optimizer with a learning rate of 0.005.

3.3. Evaluation

We evaluate the ability of the models to calibrate and summarize radiocarbon dates, that is, to estimate the true ages of radiocarbon samples and to reconstruct the true density from which they are drawn (Supplement A, Evaluation). We measure age estimation performance using a Monte Carlo estimate of the continuous ranked probability score (CRPS)

$$\mathrm{CRPS} = \frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{E}\left|\hat{t}_i - t_i\right| - \frac{1}{2}\mathbb{E}\left|\hat{t}_i - \hat{t}_i'\right|\right] \tag{12}$$

with $\hat{t}_i$ drawn from i's posterior given its fraction modern F_i and $\hat{t}_i'$ an independent draw from the same posterior. The first term is the expected absolute error, the second the expected spread in samples. Subtracting the latter from the former removes the implicit reward for being over-

confident in age estimates. For evaluation, we use 1000 posterior draws. For interpretability, we define an oracle model that knows the true age distribution $p(t)$ perfectly, and have it do posterior age estimation with the measurement matrix $\mathbf{M}$. The idea here is to define a theoretical baseline CRPS_o on attainable error in age estimation in the face of measurement and calibration curve uncertainty while controlling for all other sources of uncertainty. The exact comparison is

$$\text{Recovery} = \frac{\text{CRPS} - \text{CRPS}_o}{\text{CRPS}_o} \tag{13}$$

showing the model's proportional excess error over the oracle, with zero indicating parity. The oracle is optimal in expectation, so the recovery should be non-negative on average, though sampling variability may lead some values to dip below zero.

To measure density estimation performance, we calculate total variation distance for the discrete evaluation of the density $\mathbf{v} \approx p(t)$ and its estimate $\hat{\mathbf{v}}$ over the age grid, which is just one-half the sum of the absolute differences at each year $j = 1, ..., m$ for the normalized distributions, or

$$\text{TVD} = 0.5 \sum_{j=1}^{m} \left| v_j - \hat{v}_j \right| \tag{14}$$

Both metrics are recorded for all models over all 2268 simulations.

We conduct crude benchmarks as well using the same simulation architecture across different sample sizes n ranging from 500 to 5000 and $m = \{500, 2500, 12500\}$ years, so different sizes of the measurement matrix $\mathbf{M}$ (Supplement C). The number of clusters is held constant at $K = 6$ and the distribution for the mixture is restricted to a Gaussian since neither is a meaningful constraint on compute time. Total elapsed time depends on how long each model is allowed to run, so the comparison is meaningful only if each runs long enough to converge. As discussed above, we use fixed run lengths: 4,000 iterations for BGMM, 50,000 for DPMM, and 512 epochs for NetC14. We check convergence under less demanding conditions than those used in the benchmarks ($K = 3$, $m = 2500$, $n = \{50, 250, 1250\}$), running four chains of each MCMC model and computing the rank-normalized split $\hat{R}$ (Vehtari et al., 2021) on label-invariant quantities.

Table 1: Evaluation results across all models, sample sizes, and number of clusters. Parentheses show 95% intervals.

Model	CRPS-Recovery	CRPS	TVD	Elapsed (s)
BGMM	0.050 (-0.006, 1.145)	41.596 (9.779, 118.909)	0.206 (0.018, 0.659)	57.190 (28.046, 374.507)
CKDE	0.104 (-0.008, 3.959)	44.652 (16.306, 125.313)	0.306 (0.058, 0.663)	1.390 (0.261, 6.192)
DPMM	0.055 (-0.005, 1.417)	42.139 (9.908, 126.182)	0.205 (0.024, 0.640)	293.663 (41.077, 690.830)
NetC14	0.051 (-0.009, 1.361)	41.015 (11.927, 117.137)	0.196 (0.029, 0.650)	6.231 (3.220, 23.655)
SPD	0.156 (-0.002, 4.761)	48.445 (20.542, 130.510)	0.263 (0.064, 0.756)	0.028 (0.001, 0.343)

3.4. Archaeological Application

We apply all models to two real world datasets used in Heaton (2022) and Bronk Ramsey (2017), including dates from Iron Age Ireland (Armit et al., 2014) and Paleoindian North America (Buchanan et al., 2008). Armit et al. (2014) use a substantial radiocarbon dataset ($n = 2023$) from Ireland to reconstruct the population history of the island during the transition from the Bronze Age to the early Iron Age, from roughly 3200 to 2500 ybp. Comparing their population reconstruction to climate reconstructions over the same interval allows them to evaluate the hypothesis that a climate downturn led to population decline leading into the Iron Age. Buchanan et al. (2008) use a smaller sample ($n = 628$) to reconstruct the population history of Paleoindians in North America at the onset of a prolonged drying period known as the Younger Dryas, from 15000 to 9000 ybp. The population reconstruction allows them to evaluate the hypothesis that large low-density extraterrestrial objects impacted North America around 12900 ybp, causing the Younger Dryas and subsequent population decline among North American megafauna and Paleoindians (Firestone et al., 2007). Both papers apply SPD to the measurement matrix $\mathbf{M}$ to estimate $p(t)$, using it as a proxy for population size over time. For comparative purposes, we apply all models to these data and evaluate differences graphically (Supplement A, Real Data).

4. Results

Table 1 reports general results of the simulations showing that the mixture models are roughly equal in their ability to calibrate and summarize radiocarbon dates. The panels of Figure 2 show error metrics for models fit to simulated fraction modern data across number of samples n, number of years m, number of clusters K, and type of distribution f in each simulation. In each of these figures, the origin represents the oracle. Values along the x-axis give the distance of the estimated density from the true density, while values along the y-axis give the calibration error in excess of what is achievable with the true density as a prior. Consequently, the x-intercept at $y = 0$ is the point at which a model recovers the oracle's age estimates perfectly and will in general be strictly positive, showing what remains of the error in density estimation when uncertainty in the sample ages is removed. The y-intercept, on the other hand, should in theory fall precisely on the origin since $x = 0$ implies that the true distribution has been estimated perfectly, which is the epistemic state of the oracle and the baseline for recovery.

Figure 2 (A) shows that the three mixture models are roughly equivalent with respect to sample size, approaching the origin as n increases, thus improving along both axes. CKDE moves in a similar direction but with less improvement. The SPD appears to be insensitive to sample size when it comes to calibration performance, but does increase in summary performance. Figure 2 (B) shows that all models converge on the oracle as the number of years m in the interval increases. The SPD and CKDE both underperform at narrower intervals.

Figure 2 (C) shows that as the number of clusters in the simulation increases, the ability of the mixture models to reconstruct the true density diminishes, while performance in age estimation remains stable. SPD is largely insensitive to the number of clusters when it comes to summarizing dates, but does improve noticeably in calibration performance. CKDE exhibits a dramatic swing in summary performance across number of clusters, but is mostly stationary on calibration performance. It does perform almost as well as the mixture models along that axis.

Figure 2 (D) shows that the mixture models all do more or less equally well when it comes to data drawn from Gaussian mixtures and all do equally worse when it comes to data drawn from other component distributions. SPD and CKDE exhibit similar trajectories while performing generally worse than the mixture models. Note that summary scores are more variable than calibration performance. The one exception to this is the SPD, which improves significantly in calibration

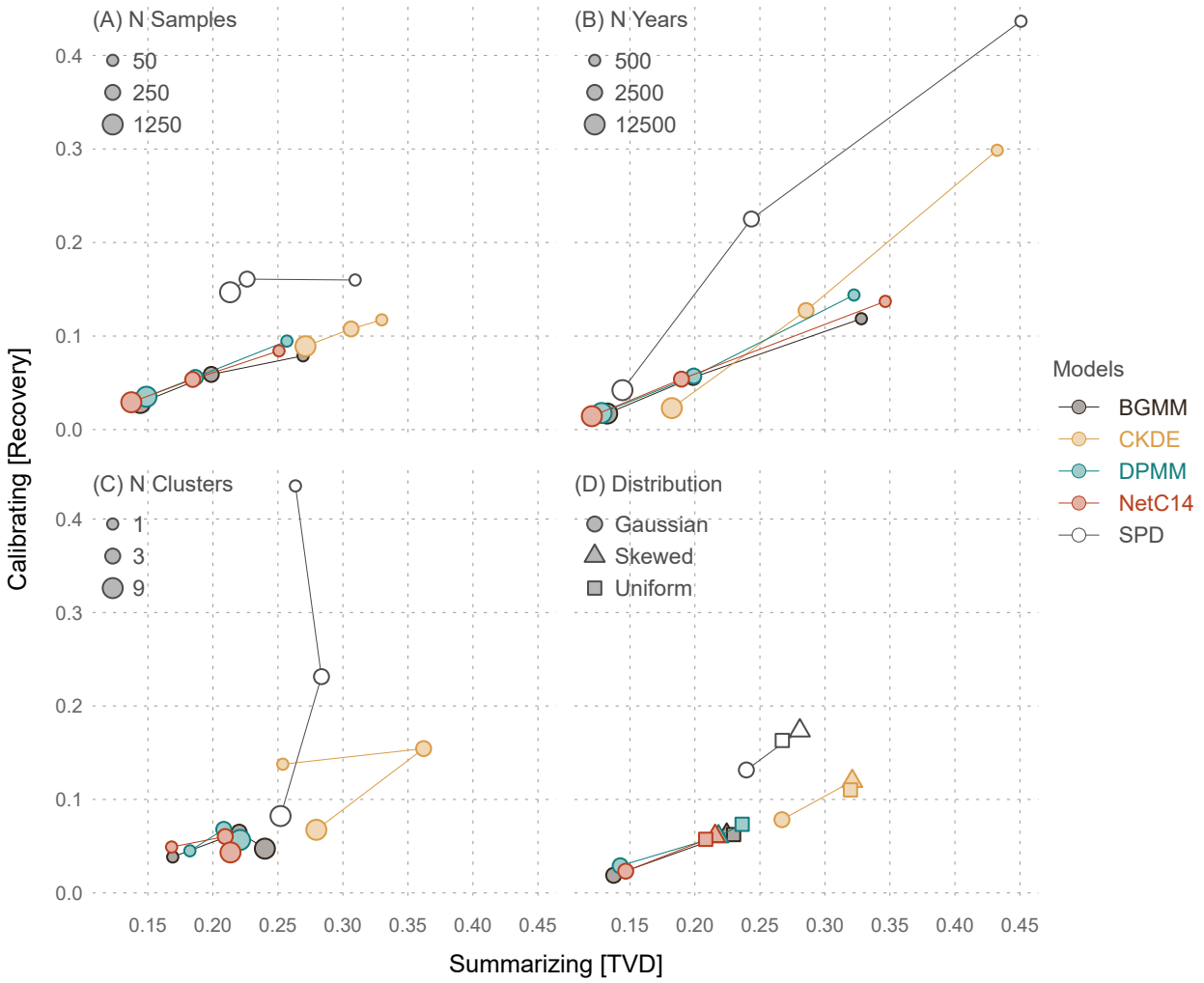


Figure 2: Evaluation results for each model by (A) sample size, (B) number of years, (C) number of clusters, and (D) distribution type. Recovery of CRPS on the y-axis shows calibration performance. TVD on the x-axis shows summary performance. Coordinates are median values across both dimensions.

performance as the number of clusters making up the true mixture density increases. Nevertheless, SPD performs considerably worse than the other methods on both dimensions.

Figure 3 shows that the mixture models all more or less converge on the true number of components. However, there is some tendency to underestimate the number of components, particularly for small samples. The BGMM also has a slight tendency to underestimate the number of components across all distributions. The DPMM appears to be relatively consistent in its behavior across sample sizes and distributions. Finally, NetC14 shows a tendency to reach for additional components as sample size increases, particularly for the skewed mixtures.

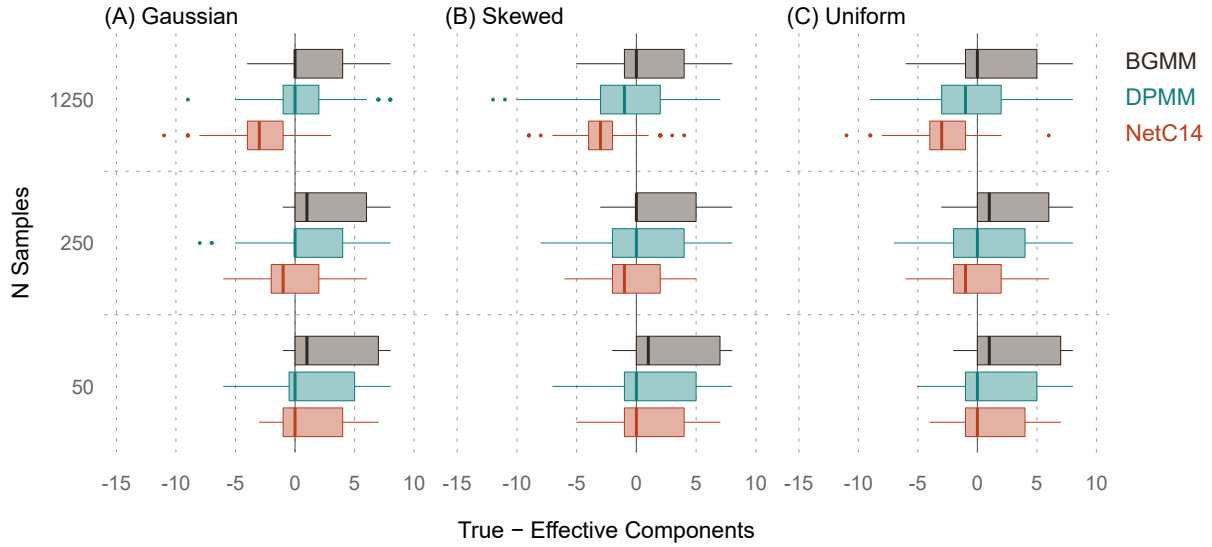


Figure 3: Distribution of the error in the number of mixture components (true number of clusters minus effective components) estimated by BGMM, DPMM, and NetC14 at each sample size for simulations with (A) Gaussian, (B) skewed, and (C) uniform components, where positive values indicate too few components and negative values too many.

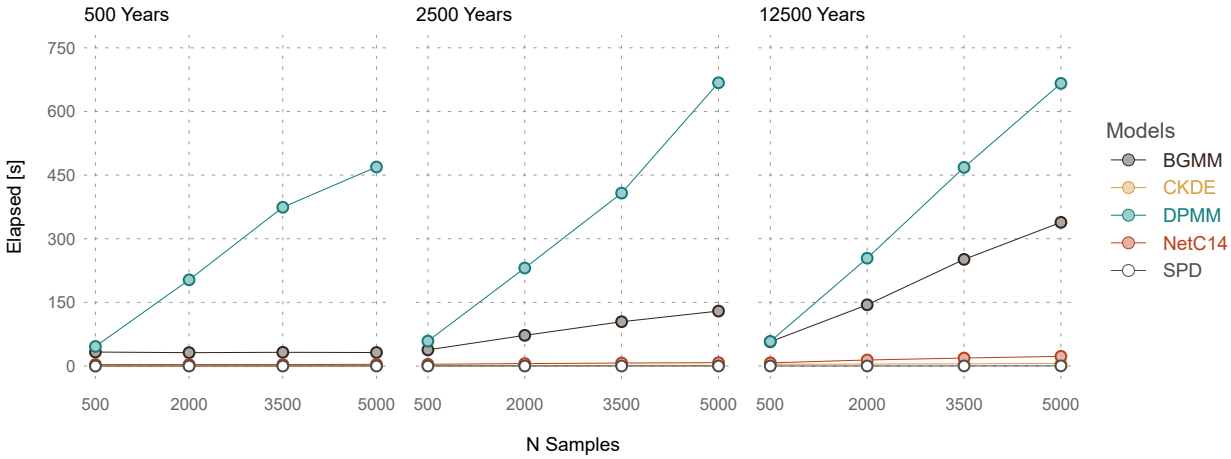


Figure 4: Median elapsed time for model training across different sample sizes n and number of years m, so that each panel shows the size of the $n \times m$ matrix $\mathbf{M}$.

While the mixture models are more or less equally competitive in age and density estimation, compute times vary substantially, as shown in Figure 4. The SPD is clearly fastest, followed closely by CKDE, with median compute times of 0.03 and 1.39s respectively for $m = \{500, 2500, 12500\}$ and $K = 6$, with n ranging between 500 and 5000. The next fastest is NetC14 at a median compute time of 6.23s, followed by the torch implemented BGMM at 57.19s and DPMM at 293.66s. BGMM

and NetC14 also appear to scale much better than the DPMM over different sample sizes, though the BGMM struggles with wider intervals. On convergence, median $\hat{R}$ fell below 1.01 for both MCMC models, and NetC14′s training loss plateaued well before the final epoch (Supplement A, Convergence Test). Finally, applications to real world datasets show general agreement across models regarding the shape of the age distribution that the samples are drawn from, as shown in Figure 5 and Figure 6. The main exceptions are, one, that the fully Bayesian mixture models place probability spikes at several places and, two, that the CKDE tends to smooth out the distribution.

5. Discussion

In simulations, NetC14 achieves parity with existing mixture models. Across number of samples n, number of years m, number of clusters K, and distributions f, the mixture models are more or less in lockstep. It is informative, therefore, to look at how performance for all models varies across scenarios, beginning with the role of the prior. For this, panel B in Figure 2 is instructive, as it shows all models converging toward the oracle in both objectives as the interval widens.

Consider first recovery, which measures what is lost by not knowing the true age distribution $p(t)$. Over 500-year intervals, SPD's median excess error over the oracle is roughly 44%, compared with 12-14% for the mixture models, but over 12,500-year intervals, where components are proportionally wider, these fall to 4% and 1-2%, respectively. This suggests that when components are wide relative to the uncertainty of individual dates, the prior is nearly flat over each date's likelihood, so the likelihood dominates the age posterior and the prior adds little. Conversely, when components are narrow, the prior contributes substantially more to age estimation, as in Figure 1.

Similar reasoning applies to summarizing dates. Measurement and calibration uncertainty spread each sample's probability mass over decades or even centuries. When components are narrow relative to that spread, much of the mass falls outside the component a sample came from, so estimates of $p(t)$ that sum the rows of $\mathbf{M}$ directly, like SPD and CKDE, are badly distorted, hence higher TVD values. The mixture models instead treat that spread as something to explain, placing components where samples overlap, which pulls each sample's mass back toward the component that generated it. By contrast, when components are relatively wide, little mass leaves components in the first place, so there is little distortion to correct and SPD and CKDE approach parity with the mixture models. Even so, no model reaches the origin along the x-axis. The remaining error

presumably reflects the familiar difficulty of estimating a density from a finite sample, with small differences between models reflecting their own limitations.

The same logic may also explain why increasing sample size contributes to better performance for mixture models across both objectives, but to SPD only in summary performance. More samples keeps more of the average mass around the components but does nothing to help with independent calibration. The fact that the SPD exhibits dramatic improvement in calibration as the number of clusters increases is probably also owing to the fact that the true density becomes increasingly flat where it overlaps the spread of each sample. As for the skewed and uniform distributions, the answer there is straightforward. All of the models have continuous, unbounded support, but the skewed and uniform are bounded (meaning they can be exactly zero over some range of years), so the models are forced to move some amount of mass outside the range of the components. The uniform also has discontinuous jump points that Gaussians would struggle to represent.

All three mixture models estimate similar component counts across small samples and distributions, as shown in Figure 3, but NetC14 and the DPMM overestimate the number of components as the sample grows, though the effect is more pronounced for NetC14. This is owing to two features of the models, their initialization and their priors. The BGMM's slight but systematic undercount is probably owing to the BIC penalty on complexity in the initialization step being too harsh, assuming that each noisy date carries the same amount of information as one that is exact. The DPMM introduces new components at a rate of roughly $\alpha_d \log n$ and is known to introduce small, ephemeral clusters into the mixture (Ji et al., 2010; Miller & Harrison, 2014). NetC14 introduces components owing to both its initialization and its priors. The starting number of components grows as $\sqrt[3]{n}$, the log-likelihood ℓ grows linearly in n, and the precision penalty grows with the precision of active components, scaling by the length of the time window m rather than n. So as n grows, the data can afford more, and narrower, active components, and SparseMax prunes only those whose logits fall below its threshold.

While estimating the number of components K is a well-known problem for mixture models (Cai et al., 2021), how much it matters depends on the questions we want to ask. For radiocarbon dating in archaeology, and in the study of human behavior more generally, estimating K is arguably of little relevance. The quantity of most scientific interest is instead the density $p(t)$, as it provides a relative measure of aggregate demand for resources and, thus, of the level of competition in the economy. Next in importance are the ages themselves, as they measure variability within the

population. If we care about the K components and their parameters θ at all, then, it is only because they provide an efficient, compressed representation of the signal $p(t)$. Indeed, it is precisely because $p(t)$ admits such compression that it counts as a signal at all (Chaitin, 1975; Dennett, 1991).

The catch, of course, is that compression can introduce artifacts - or real patterns, depending on who you ask. When applied to the Iron Age Ireland dataset in Figure 5, for example, both fully Bayesian mixture models place a large probability spike at 1,550 ybp, and the DPMM places further spikes in the Paleoindian North America dataset at around 9,100 and 9,500 ybp in Figure 6. NetC14 places no comparable spikes, and its components have broadly similar widths. This is owing to its relatively severe precision penalty, which puts a floor of roughly $2\gamma m/n_k$ on a component's variance. A narrow, sparsely populated component is, as a consequence, likely to be absorbed into a neighboring component or pruned by SparseMax. The BGMM and DPMM place a more relaxed penalty on the variance. Under their Gamma priors, a handful of dates can support a very narrow component provided they are tightly clustered. Because the true age distribution of real data is unknown, however, we cannot say which behavior is correct here. The spikes may reflect real, short-lived episodes of activity, or they may be artifacts of the calibration curve and the priors.

While all three mixture models perform more or less equally well at age and density estimation, they differ in compute efficiency, with NetC14 being much faster. Because NetC14 uses penalized maximum likelihood, it only has to compute the log likelihood once per epoch. The BGMM also marginalizes over calendar ages by evaluating the mixture density across the measurement matrix $\mathbf{M}$. However, it has to repeat that evaluation, along with its gradient, at every jump of the leapfrog integrator (12 by default) in every MCMC iteration. The DPMM never builds $\mathbf{M}$. Instead, it treats each sample's calendar age as a latent variable and updates it with a slice sampler (Neal, 2003), evaluating the calibration curve only at proposed ages. It also reassigns each sample to a cluster through the Pólya urn. Each iteration is, therefore, nearly independent of m. That is why the BGMM slows as the interval widens while the DPMM mostly slows with sample size. The DPMM is likely the slowest overall because its updates run sequentially, one sample at a time, while the BGMM's likelihood is a single vectorized matrix product. It is also worth noting that NetC14 was severely handicapped in these benchmarks, since it does not use torch's parallelism or GPU acceleration. Furthermore, larger samples, longer intervals, and more clusters should, if anything,

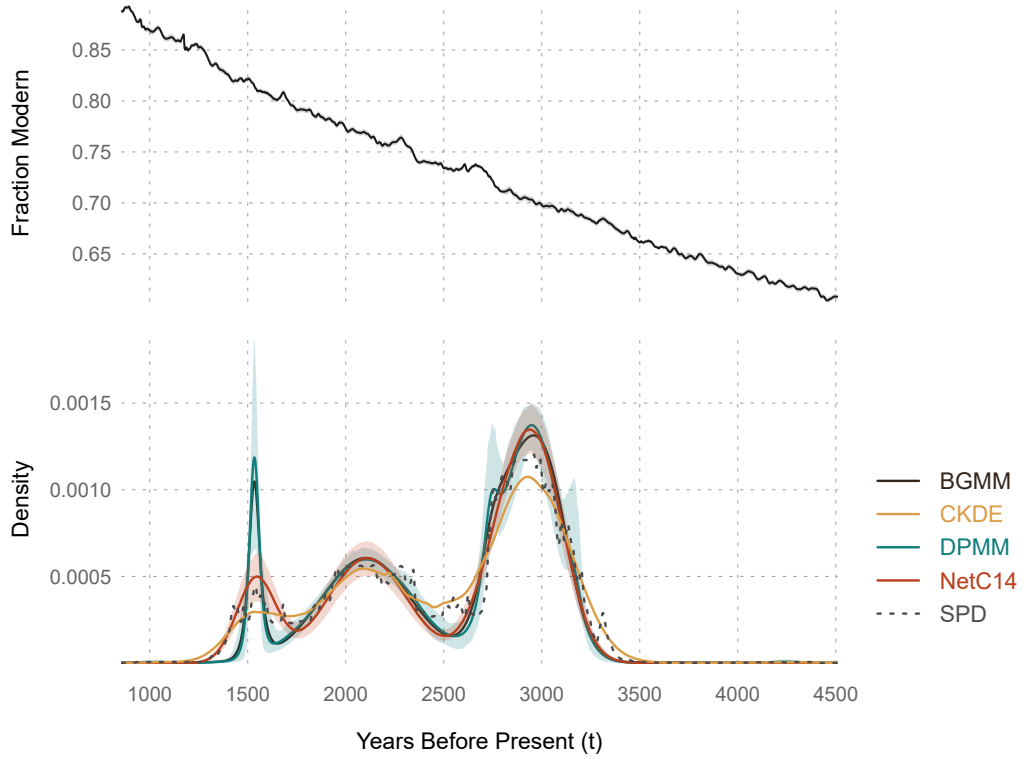


Figure 5: Iron Age Ireland. Methods applied to radiocarbon dates from Armit et al. (2014).

require more MCMC iterations to converge, so the benchmarks likely understate the difference in compute time between NetC14 and the fully Bayesian mixture models.

The analysis serves as a proof of concept for the use of neural networks in radiocarbon dating. To that end, we have severely restricted NetC14 to give a sense of the general architecture and method, rather than going all in on all the available features neural networks currently offer. In fact, we probably could have fit the same model with simple Expectation Maximization rather than using the Adam optimizer. To what extent then is NetC14 even a neural network? Our answer to this is twofold. First, we see the main contribution of NetC14 to be in the specification of the loss function, which is a simple implementation of equation 14 in Price et al. (2021) (Equation 8 above) with an additional precision penalty to prevent components from collapsing around a handful of dates (Equation 11). The ease with which one can implement a custom loss function and train a model with back propagation and gradient descent is easily one of the most powerful features of neural networks. Second, NetC14 is by the letter a neural network. It is just a shallow or one-layer neural network, that layer being the parameter vector θ. That is not even remotely a limitation, however, as neural networks are almost infinitely composable. Consequently, one should think of

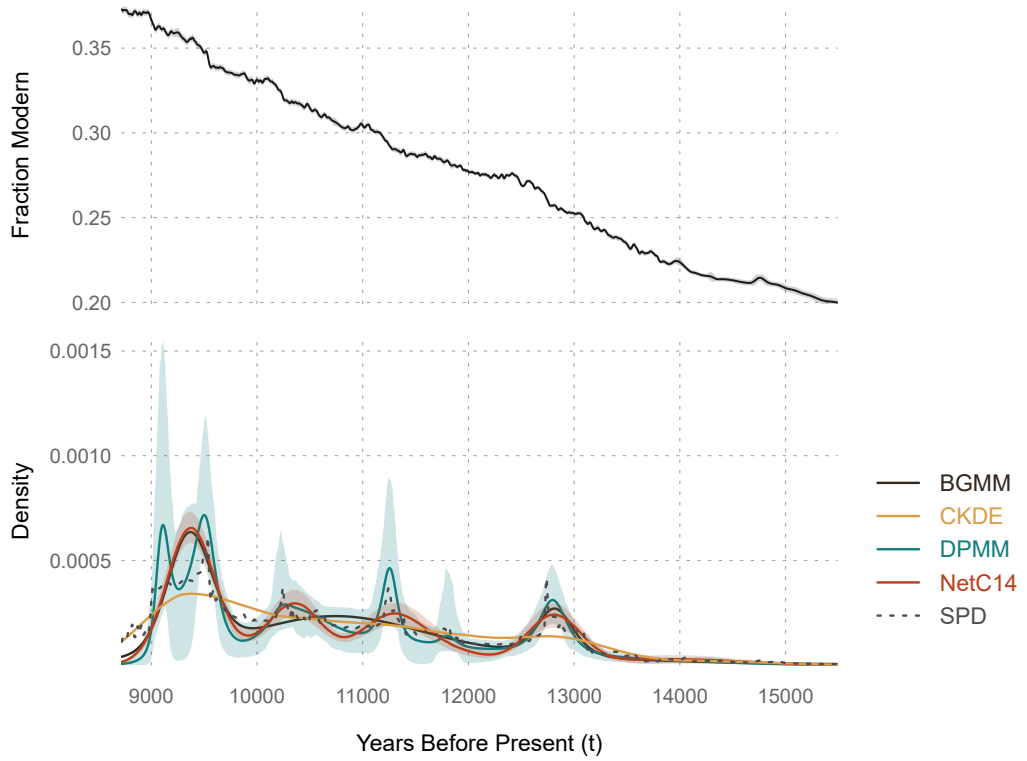


Figure 6: Paleoindian North America. Methods applied to radiocarbon dates from Buchanan et al. (2008).

NetC14's single layer as the last layer in a potentially very deep neural network composed of many interesting and powerful modules.

This is the more compelling difference between NetC14 and the fully Bayesian mixture models, the ease with which it can be extended. For example, because θ is simply the output of a network layer, the mixture parameters can be made functions of additional sample-level data x, $\theta(x) = \{w_k(x), \mu_k(x), \sigma_k(x)\}$, so that they vary across radiocarbon samples $i = 1, ..., n$. This is known as a Mixture Density Network (MDN) (Bishop, 1994). Covariate-dependent mixtures are not unique to neural networks. Dependent Dirichlet processes and regression on mixture weights are well established in the Bayesian literature (Wade & Inácio, 2025). However, gradient-based training makes it comparatively straightforward to condition on high-dimensional inputs and to reuse existing architectures for them in a way that scales. Examples include encoders for ceramic or projectile point assemblage matrices and convolutional networks for imagery or Lidar over sample contexts.

Such extensions would let the model output artifact production profiles, which have long been of interest to archaeologists (Ford, 1938; Lipo et al., 2015). They would also allow indirect dating, that is, predicting occupation sequences for sites and regions that lack direct dates but have diagnostics available (Crema, 2025; Ortman, 2016). And they would allow archaeological hypotheses about population change to be tested within a single model by including presumed causal variables. But naturally, caveats apply. First, additional data do not remove the ambiguity that plateaus in the calibration curve introduce, because that ambiguity is in the likelihood. Covariates can sharpen age estimates only insofar as they carry independent information about age, as stratigraphic or contextual priors do in a Bayesian model. Second, typical radiocarbon datasets contain hundreds of dates, sometimes thousands, and in rare cases tens of thousands. Conditioning on high-dimensional inputs will therefore require strong regularization or pretrained components, and more speculative inputs, such as text embeddings of the archaeological literature, would face this problem most acutely.

All of the methods evaluated here assume that radiocarbon dates are representative of the true age distribution. In our simulations they are, but in reality many intervening forces may bias the sample (Attenbrow & Hiscock, 2015; Contreras & Codding, 2024; Contreras & Meadows, 2014; Crema, 2022; Rick, 1987). From the moment an organic sample is deposited, it begins to break down, with its rate of decay and chances of surviving to the present depending on both the sample itself and its environment (Contreras & Codding, 2024; Surovell et al., 2009). And a sample that does survive is not sampled at random either. Dating is costly and thus biased toward the interests of well-funded researchers, and regions that attract more archaeologists tend to produce larger samples (Codding et al., 2024). Existing corrections, such as the Surovell correction for taphonomic decay (Surovell et al., 2009) and stratified sampling for research intensity (e.g., `rcarbon::thinDates`), could be incorporated into NetC14, for instance through weighted or stratified mini-batch sampling. The literature on debiasing neural networks (Wang & Liu, 2023) may also offer additional tools for this persistent challenge.

Archaeologists have also long recognized that population reconstructions based on radiocarbon dates often imply implausible demographic rates (Brown, 2015; Contreras & Meadows, 2014). Because the estimated density can take whatever shape the data suggest, reconstructions can show growth or decline over short intervals far beyond what human populations can sustain, particularly where the calibration curve introduces artificial peaks and troughs. Bayesian methods address this

by constraining reconstructions with explicit growth models (Crema & Shoda, 2021; Wilson et al., 2026). A neural network can pursue the same end by imposing demographic expectations as penalties in the loss function, as physics-informed neural networks do physical processes (Raissi et al., 2019). Whether it offers practical advantages over explicit Bayesian growth models remains to be tested, but it fits naturally within NetC14′s framework, given how easily its loss can be modified.

6. Conclusion

Here we systematically compare the major extant methods and introduce NetC14, a novel application of neural networks using regularized Maximum Likelihood to simultaneously estimate the age of an archaeological event and the distribution of many dated events. The results show that NetC14 performs as well as the state of the art in Bayesian radiocarbon dating methods. The computational advantages of NetC14, plus the promise to expand the application to include additional covariates and network architectures allowing to address archaeological questions, should improve the uptake of these approaches while also expanding the potential to explain past human behavior.

Bibliography

Armit, I., Swindles, G. T., Becker, K., Plunkett, G., & Blaauw, M. (2014). Rapid climate change did not cause population collapse at the end of the European Bronze Age. *Proceedings of the National Academy of Sciences*, *111*(48), 17045–17049. https://doi.org/10.1073/pnas.1408028111

Attenbrow, V., & Hiscock, P. (2015). Dates and demography: Are radiometric dates a robust proxy for long-term prehistoric demographic change?. *Archaeology in Oceania*, *50*(S1), 30–36. https://doi.org/10.1002/arco.5052

Bayliss, A. (2009). Rolling out revolution: Using radiocarbon dating in archaeology. *Radiocarbon*, *51*(1), 123–147. https://doi.org/10.1017/S0033822200033750

Bishop, C. M. (1994). *Mixture Density Networks* (Technical Report NCRG/94/004; pp. 1–25). Aston University Neural Computing Research Group. https://publications.aston.ac.uk/id/eprint/373/1/NCRG_94_004.pdf

Blockley, S. P. E., Donahue, R. E., & Pollard, A. M. (2000). Radiocarbon calibration and Late Glacial occupation in northwest Europe. *Antiquity*, *74*(283), 112–119. https://doi.org/10.1017/S0003598X00066199

Bronk Ramsey, C. (2008). Radiocarbon dating: Revolutions in understanding. *Archaeometry*, *50*(2), 249–275. https://doi.org/10.1111/j.1475-4754.2008.00394.x

Bronk Ramsey, C. (2017). Methods for Summarizing Radiocarbon Datasets. *Radiocarbon*, *59*(6), 1809–1833. https://doi.org/10.1017/RDC.2017.108

Brown, W. A. (2015). Through a filter, darkly: population size estimation, systematic error, and random error in radiocarbon-supported demographic temporal frequency analysis. *Journal of Archaeological Science*, *53*, 133–147. https://doi.org/10.1016/j.jas.2014.10.013

Brown, W. A. (2017). The past and future of growth rate estimation in demographic temporal frequency analysis: Biodemographic interpretability and the ascendance of dynamic growth models. *Journal of Archaeological Science*, *80*, 96–108. https://doi.org/10.1016/j.jas.2017.02.003

Buchanan, B., Collard, M., & Edinborough, K. (2008). Paleoindian demography and the extraterrestrial impact hypothesis. *Proceedings of the National Academy of Sciences*, *105*(33), 11651–11654. https://doi.org/10.1073/pnas.0803762105

Cai, D., Campbell, T., & Broderick, T. (2021). Finite Mixture Models Do Not Reliably Learn the Number of Components. In M. Meila & T. Zhang (Eds.), *Proceedings of the 38th International Conference on Machine Learning: Vol. 139. Proceedings of the 38th International Conference on Machine Learning*.

Carleton, W. C., & Groucutt, H. S. (2021). Sum things are not what they seem: Problems with point-wise interpretations and quantitative analyses of proxies based on aggregated radiocarbon dates. *The Holocene*, *31*(4), 630–643.

Chaitin, G. J. (1975). Randomness and Mathematical Proof. *Scientific American*, *232*(5), 47–52.

Chen, J., Tan, X., & Zhang, R. (2008). Inference for Normal Mixtures in Mean and Variance. *Statistica Sinica*, *18*(2), 443–465.

Ciuperca, G., Ridolfi, A., & Idier, J. (2003). Penalized Maximum Likelihood Estimator for Normal Mixtures. *Scandinavian Journal of Statistics*, *30*(1), 45–59. https://doi.org/10.1111/1467-9469.00317

Codding, B. F., Meyer, J., Brewer, S. C., Kelly, R. L., & Jones, T. L. (2024). Do counts of radiocarbon-dated archaeological sites reflect human population density? A preliminary empirical validation examining spatial variation across late Holocene California. *Radiocarbon*, *66*(4), 610–623. https://doi.org/10.1017/RDC.2024.81

Contreras, D. A., & Codding, B. F. (2024). Landscape taphonomy predictably complicates demographic reconstruction. *Journal of Archaeological Method and Theory*, *31*(3), 1102–1128. https://doi.org/10.1007/s10816-023-09634-5

Contreras, D. A., & Meadows, J. (2014). Summed radiocarbon calibrations as a population proxy: A critical evaluation using a realistic simulation approach. *Journal of Archaeological Science*, *52*, 591–608. https://doi.org/10.1016/j.jas.2014.05.030

Crema, E. R. (2022). Statistical inference of prehistoric demography from frequency distributions of radiocarbon dates: A review and a guide for the perplexed. *Journal of Archaeological Method and Theory*, *29*(4), 1387–1418. https://doi.org/10.1007/s10816-022-09559-5

Crema, E. R. (2025). A Bayesian alternative for aoristic analyses in archaeology. *Archaeometry*, *67*(S1), 7–30. https://doi.org/10.1111/arcm.12984

Crema, E. R., & Shoda, S. (2021). A Bayesian approach for fitting and comparing demographic growth models of radiocarbon dates: A case study on the Jomon-Yayoi transition in Kyushu (Japan). *PLOS ONE*, *16*(5), e0251695. https://doi.org/10.1371/journal.pone.0251695

Day, N. E. (1969). Estimating the Components of a Mixture of Normal Distributions. *Biometrika*, *56*(3), 463–474. https://doi.org/10.2307/2334652

Dennett, D. C. (1991). Real Patterns. *The Journal of Philosophy*, *88*(1), 27–51. https://doi.org/10.2307/2027085

Falbel, D., & Luraschi, J. (2026). *torch: Tensors and neural networks with 'GPU' acceleration* [Technical report]. https://torch.mlverse.org/docs

Firestone, R. B., West, A., Kennett, J. P., Becker, L., Bunch, T. E., Revay, Z. S., Schultz, P. H., Belgya, T., Kennett, D. J., Erlandson, J. M., Dickenson, O. J., Goodyear, A. C., Harris, R. S., Howard, G. A., Kloosterman, J. B., Lechler, P., Mayewski, P. A., Montgomery, J., Poreda, R., … Wolbach, W. S. (2007). Evidence for an extraterrestrial impact 12,900 years ago that contributed to the megafaunal extinctions and the Younger Dryas cooling. *Proceedings of the National Academy of Sciences*, *104*(41), 16016–16021. https://doi.org/10.1073/pnas.0706977104

Ford, J. A. (1938). A chronological method applicable to the Southeast. *American Antiquity*, *3*(3), 260–264. https://doi.org/10.2307/275264

Graves, A. (2011). Practical Variational Inference for Neural Networks. In J. Shawe-Taylor, R. Zemel, P. Bartlett, F. Pereira, & K. Weinberger (Eds.), *Advances in Neural Information Processing Systems: Vol. 24. Advances in Neural Information Processing Systems*.

Hajdas, I., Ascough, P., Garnett, M. H., Fallon, S. J., Pearson, C. L., Quarta, G., Spalding, K. L., Yamaguchi, H., & Yoneda, M. (2021). Radiocarbon dating. *Nature Reviews Methods Primers*, *1*(1), 62. https://doi.org/10.1038/s43586-021-00058-7

Heaton, T. J., Bard, E., Bayliss, A., Blaauw, M., Bronk Ramsey, C., Reimer, P. J., Turney, C. S. M., & Usoskin, I. (2024). Extreme solar storms and the quest for exact dating with radiocarbon. *Nature*, *633*(8029), 306–317. https://doi.org/10.1038/s41586-024-07679-4

Heaton, T. J., Blaauw, M., Blackwell, P. G., Bronk Ramsey, C., Reimer, P. J., & Scott, E. M. (2020). The IntCal20 approach to radiocarbon calibration curve construction: A new methodology using Bayesian splines and errors-in-variables. *Radiocarbon*, *62*(4), 821–863. https://doi.org/10.1017/RDC.2020.46

Heaton, T. J. (2022). Non-parametric calibration of multiple related radiocarbon determinations and their calendar age summarisation. *Journal of the Royal Statistical Society Series C: Applied Statistics*, *71*(5), 1918–1956. https://doi.org/10.1111/rssc.12599

Hoffman, M. D., & Gelman, A. (2014). The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, *15*(1), 1593–1623.

Ji, Y., Lin, T., & Zha, H. (2010). CDP Mixture Models for Data Clustering. *2010 20th International Conference on Pattern Recognition*, 637–640. https://doi.org/10.1109/ICPR.2010.161

Kingma, D. P., & Ba, J. (2017). *Adam: A method for stochastic optimization*. https://arxiv.org/abs/1412.6980

Li, J., & Barron, A. (1999). *Mixture density estimation* (S. Solla, T. Leen, & K. Müller, Eds.; Vol. 12). MIT Press. https://proceedings.neurips.cc/paper_files/paper/1999/file/a0f3601dc682036423013a5d965db9aa-Paper.pdf

Libby, W. F. (1946). Atmospheric helium three and radiocarbon from cosmic radiation. *Physical Review*, *69*(11–12), 671–672. https://doi.org/10.1103/PhysRev.69.671.2

Lipo, C. P., Madsen, M. E., & Dunnell, R. C. (2015). A theoretically-sufficient and computationally-practical technique for deterministic frequency seriation. *PLOS ONE*, *10*(4), e0124942. https://doi.org/10.1371/journal.pone.0124942

Martins, A., & Astudillo, R. (2016). From Softmax to Sparsemax: A Sparse Model of Attention and Multi-Label Classification. In M. F. Balcan & K. Q. Weinberger (Eds.), *Proceedings of The 33rd International Conference on Machine Learning: Vol. 48. Proceedings of The 33rd International Conference on Machine Learning*.

McLachlan, G. J., & Peel, D. (2000). *Finite Mixture Models*. Wiley.

McLachlan, G. J., Lee, S. X., & Rathnayake, S. I. (2019). Finite mixture models. *Annual Review of Statistics and Its Application*, *6*(Volume 6, 2019), 355–378. https://doi.org/https://doi.org/10.1146/annurev-statistics-031017-100325

McLaughlin, T. R. (2019). On applications of space–time modelling with open-source 14C age calibration. *Journal of Archaeological Method and Theory*, *26*(2), 479–501. https://doi.org/10.1007/s10816-018-9381-3

McNichol, A. P., Jull, A. J. T., & Burr, G. S. (2001). Converting AMS data to radiocarbon values: Considerations and conventions. *Radiocarbon*, *43*(2A), 313–320. https://doi.org/10.1017/S0033822200038169

Miller, J. W., & Harrison, M. T. (2014). Inconsistency of Pitman-Yor Process Mixtures for the Number of Components. *Journal of Machine Learning Research*, *15*(96), 3333–3370.

Neal, R. M. (2000). Markov chain sampling methods for Dirichlet process mixture models. *Journal of Computational and Graphical Statistics*, *9*(2), 249–265. https://doi.org/10.1080/10618600.2000.10474879

Neal, R. M. (2003). Slice sampling. *The Annals of Statistics*, *31*(3), 705–767. https://doi.org/10.1214/aos/1056562461

Ortman, S. G. (2016). Uniform Probability Density Analysis and Population History in the Northern Rio Grande. *Journal of Archaeological Method and Theory*, *23*(1), 95–126. https://doi.org/10.1007/s10816-014-9227-6

Price, M. H., Capriles, J. M., Hoggarth, J. A., Bocinsky, R. K., Ebert, C. E., & Jones, J. H. (2021). End-to-end Bayesian analysis for summarizing sets of radiocarbon dates. *Journal of Archaeological Science*, *135*, 105473. https://doi.org/10.1016/j.jas.2021.105473

R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. https://www.r-project.org/

Raissi, M., Perdikaris, P., & Karniadakis, G. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, *378*, 686–707. https://doi.org/10.1016/j.jcp.2018.10.045

Reimer, P. J., Austin, W. E. N., Bard, E., Bayliss, A., Blackwell, P. G., Bronk Ramsey, C., Butzin, M., Cheng, H., Edwards, R. L., Friedrich, M., Grootes, P. M., Guilderson, T. P., Hajdas, I., Heaton, T. J., Hogg, A. G., Hughen, K. A., Kromer, B., Manning, S. W., Muscheler, R., … Talamo, S. (2020). The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0–55 cal kBP). *Radiocarbon*, *62*(4), 725–757. https://doi.org/10.1017/RDC.2020.41

Rick, J. W. (1987). Dates as data: An examination of the Peruvian preceramic radiocarbon record. *American Antiquity*, *52*(1), 55–73. https://doi.org/10.2307/281060

Scott, D. W. (1979). On Optimal and Data-Based Histograms. *Biometrika*, *66*(3), 605–610. https://doi.org/10.1093/biomet/66.3.605

Surovell, T. A., Byrd Finley, J., Smith, G. M., Brantingham, P. J., & Kelly, R. (2009). Correcting temporal frequency distributions for taphonomic bias. *Journal of Archaeological Science*, *36*(8), 1715–1724. https://doi.org/10.1016/j.jas.2009.03.029

Taylor, R. E., & Bar-Yosef, O. (2016). *Radiocarbon dating: An archaeological perspective* (2nd ed.). Routledge.

Timpson, A., Barberena, R., Thomas, M. G., Méndez, C., & Manning, K. (2020). Directly Modelling Population Dynamics in the South American Arid Diagonal Using 14C Dates. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *376*(1816), 20190723. https://doi.org/10.1098/rstb.2019.0723

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian Model Evaluation Using Leave-One-out Cross-Validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432. https://doi.org/10.1007/s11222-016-9696-4

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-Normalization, Folding, and Localization: An Improved $\widehat{R}$ for Assessing Convergence of MCMC. *Bayesian Analysis*, *16*(2), 667–718. https://doi.org/10.1214/20-BA1221

Vernon, K. B. (2026,). *Kbvernon/Netc14*. https://doi.org/10.5281/zenodo.21743119

Wade, S., & Inácio, V. (2025). Bayesian Dependent Mixture Models: A Predictive Comparison and Survey. *Statistical Science*, *40*(1), 81–108. https://doi.org/10.1214/24-STS966

Wang, Y., & Liu, H. (2023). *2023 International Conference on Machine Learning and Cybernetics (ICMLC)*. 458–463. https://doi.org/10.1109/ICMLC58545.2023.10327985

Wilson, K. M., Codding, B. F., McCool, W. C., Medina, I., Vernon, K. B., & Brewer, S. C. (2026). The impact of subsistence change on demography: Theory-informed Approximate Bayesian Computation reveals 4-fold increase in carrying capacity following economic intensification. *Journal of Archaeological Science*, *189*, 106523. https://doi.org/10.1016/j.jas.2026.106523

Wood, R. (2015). From revolution to convention: The past, present and future of radiocarbon dating. *Scoping the Future of Archaeological Science: Papers in Honour of Richard Klein*, *56*, 61–72. https://doi.org/10.1016/j.jas.2015.02.019